Technosphere Collapse and Technosignature Duty Cycles

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1 Abstract

The Drake Equation's longevity term, L, is traditionally treated as a continuous span of technosignature emission, yet real civilizations may cycle between active and dormant states due to resource depletion, systemic collapse, or other disruptions. We model collapse-recovery dynamics across ten plausible futures for Earth-originating civilization using a hybrid deterministic-stochastic framework that simulates technological growth, resource depletion, and stochastic hazards over a 1000-year window. From 200 Monte Carlo simulations per scenario, we compute duty cycles (D_c) ranging from ∼0.38 to 1.00, with most scenarios exhibiting significant intermittency. We introduce an effective detectability duration, $L_{\text{eff}} = D_c T_{\text{span}}$, and an extended formulation incorporating technosignature persistence beyond active periods. We show that technological remnants from collapsed civilizations may exhibit their own ON-OFF dynamics, creating layered duty-cycle structures. Our results suggest that the Great Silence may reflect the prevalence of low-duty-cycle civilizations rather than the rarity of intelligent life. When combined with power-law longevity distributions disfavoring long-lived technosignatures, the duty-cycle framework implies a doubly diminished probability of detection. We also discuss implications for Earth's own civilizational resilience, showing how investments in knowledge preservation, genetic resource vaults, and distributed infrastructure map onto model parameters that increase the duty cycle.

2 Introduction

One of the most puzzling questions in astrobiology is whether intelligent civilizations exist elsewhere in the galaxy, and if so, why we have not yet detected them. This apparent contradiction, often referred to as the Fermi Paradox or the Great Silence, raises the question of why, in a galaxy billions of years old, we observe no clear signs of advanced extraterrestrial life. Many solutions have been proposed to explain this Great Silence [Webb, 2010]. One class of explanations suggests that intelligent life is intrinsically rare, perhaps requiring an improbable convergence of planetary, chemical, or biological conditions [Wesson, 1990]. Another contends that while intelligence may emerge, technological civilizations are typically short-lived, succumbing to self-destruction through ecological collapse, conflict, or other systemic instabilities [Vinn, 2024].

Regardless of which class of explanation is correct, the duration that a civilization—or its technosignatures—persists is the critical variable. Longevity has been recognized as the key factor in the search for technosignatures [Balbi and Ćirković, 2021]. But longevity is not a single number. A crucial and often overlooked insight is that the longevity of a technosignature can be almost entirely separable from the age of the civilization that produces it: artifacts, atmospheric signatures, and orbital debris can persist long after their creators are gone, while active transmissions may cease well before a civilization itself does [Balbi and Ćirković, 2021]. Moreover, the temporal distribution of civilizations matters as much as their abundance. The detectability requirement imposes strict constraints on the epoch of appearance and communicating lifespan, so that only a small fraction of civilizations would overlap with our observational window [Balbi, 2018]. Understanding civilizational longevity therefore requires understanding the temporal dynamics of technological activity itself.

Yet most treatments of longevity assume that once a civilization becomes technologically active, it remains so continuously until it either dies out or deliberately goes silent. This assumption is difficult to justify. Historical studies of Earth's civilizations reveal cycles of rise and fall with typical durations on the order of centuries [Shermer, 2002], and there is no strong reason to assume that extraterrestrial technological civilizations would be exempt from similar dynamics. It has been argued that planetary civilizations face a fundamental bifurcation: either "asymptotic burnout," in which superlinear growth leads to singularity-driven collapse, or "homeostatic awakening," in which a civilization deliberately curbs expansion to achieve long-term equilibrium [Wong and Bartlett, 2022]. Sustainability constraints reinforce this view: exponential expansion is likely unsustainable, so advanced societies may stabilize or collapse before spreading across the Galaxy [Haqq-Misra and Baum, 2009]. If collapse is common and recovery uncertain, then the technosphere—the totality of a civilization's technological infrastructure and output—may best be understood not as a permanent state but as an intermittent phenomenon, with concrete implications for both the search for technosignatures and for understanding our own civilizational trajectory.

If a civilization alternates between active and dormant states, its technological output is characterized not by its total lifespan but by its *duty cycle*: the fraction of time it is actively producing observable technosignatures. The concept of intermittent signals is not new to SETI—signals may be intermittent due to power conservation, planetary rotation, or targeted sequential transmissions [Gray, 2020, 2021]. But beyond signal-level intermittency, there is a deeper, civilizational-level intermittency driven by collapse-recovery dynamics. Recent work applying Lindy's law to technosignature longevities suggests that first-detected technosignatures are likely to be short-lived rather than long-lived, with power-law distributions disfavoring persistent signals [Balbi and Grimaldi, 2024], reinforcing the importance of understanding intermittency at all timescales.

In this paper, we develop a framework that connects civilizational collapse-recovery dynamics to technosignature duty cycles. Building on a recently proposed set of ten plausible futures for Earth-originating civilization spanning 1000 years [Haqq-Misra et al., 2025a], we construct a hybrid deterministic-stochastic model that simulates the long-term evolution of technological capacity and resource stocks under varying governance structures, resource regimes, and hazard exposures. Collapse and recovery cycles emerge naturally from the feedback between growth and depletion, combined with stochastic existential risk. From these simulations, we compute the average duty cycle of each scenario and derive an effective detectability duration, L_{eff}, that captures the operational reality of civilizations that switch between active and dormant states. We then explore the implications of this duty-cycle framework for the Drake Equation's longevity term L and consider what these dynamics imply for Earth's own civilizational resilience.

3 Model Description

A framework recently proposed outlines ten plausible futures for Earth-originating civilization 1000 years from now, structured around variations in governance type, resource regime, and systemic resilience [Haqq-Misra et al., 2025a]. These scenarios are grounded in Earth's present-day sociotechnical conditions and extrapolated using techniques from futures studies, including PEST analysis (Political, Economic, Social, Technological), cross-consistency assessment, and structured scenario worldbuilding. Although the framework explores plausible civilizational trajectories, it does not explicitly model the temporal dynamics of collapse and recovery, nor their implications for the persistence and detectability of technosignatures. This leaves open a crucial question: how do different sociotechnical paths shape the *ON*/*OFF* dynamics of technosignature production (and potential detection)?

To explore this, we build a dynamic model that simulates long-term evolution of technological capacity and resource stocks under varying governance structures, resource regimes, and hazard exposure. Collapse and recovery cycles emerge naturally from feedback between growth and depletion, combined with stochastic existential risk. We use this model to explore how long civilizations remain detectable, how often they go silent, and what kinds of futures might dominate the galactic landscape. From our simulations, we compute the average duty cycle of each scenario, and from this, we calculate an effective detectability duration (L_{eff}), allowing us to assess not just how long civilizations last, but how often we might catch them in the act of being detectable.

3.1 Simulation framework

We developed a hybrid deterministic-stochastic model to simulate the long-term behavior of technological civilizations, with the goal of capturing collapse-recovery dynamics and their effect on technosignature detectability. This model extends the scenario-based framework introduced by Haqq-Misra et al. [Haqq-Misra et al., 2025a], adding temporal dynamics and explicit feedback between resource depletion, technological growth, and systemic collapse. The simulation spans a fixed period of 1000 years, discretized into yearly time steps. At each step, a civilization is characterized by its technological capacity, $T(t)$, and its available resource stock, $R(t)$. Technological capacity increases linearly each year according to a fixed growth rate r, which we take directly from the values reported in Haqq-Misra et al., Table 9. Simultaneously, the available resource stock is depleted at a constant rate δ. If the resource level falls to zero (i.e., $R(t) < 0$), or if an existential risk event occurs (represented as a probabilistic hazard drawn at each time step) the civilization undergoes a collapse. Collapse represents a systemic breakdown in infrastructure, coordination, and technological continuity. It leads to a sharp reduction in technological capacity, determined by a collapse factor c_f, which defines the proportion of technology that survives the event. This loss can be interpreted as the result of disruptions to power grids, communication networks, knowledge retention, or the abandonment of complex systems.

Following collapse, the system enters a dormant recovery period of duration r_d. During this phase, the society is unable to grow or consume resources, reflecting a temporary halt in technological and economic activity. This delay allows time for institutional reorganization, population stabilization, or environmental recovery. Once the recovery period concludes, the resource stock is partially replenished to a fraction r_f of its original value R_0. This recovery fraction represents retained infrastructure, stored reserves, or ecological regeneration. A full recovery ($r_f \approx 1$) implies robust contingency planning and strong institutional memory, while lower values indicate more severe losses and diminished rebuilding capacity. This cycle of growth, collapse, dormancy, and recovery can repeat multiple times during the simulation window, depending on the fragility of the system, the frequency of stochastic hazards, and the depletion rate of resources.

Formally, let the initial conditions be $T_0 = 1$, and $R_0 \in \mathbb{R}_+$, and define the indicator function as $\chi(t) \in \{0, 1\}$, with $\chi(t) = 1$ if the system is active (not collapsed or recovering) at time t, and $\chi(t) = 0$ otherwise. The update rules for each time step are given by

$$\begin{aligned} T_{t+1} &= \chi_t \, (T_t + r) + (1 - \chi_t) \, T_t, \\ R_{t+1} &= \chi_t \, (R_t - \delta) + (1 - \chi_t) \, R_t. \end{aligned} \tag{1}$$

Equations (1)–(3) define the full dynamical cycle of growth, collapse, and recovery in the model.

We include both a deterministic and a stochastic trigger for collapse to reflect two distinct failure modes: internal depletion and external shocks. Collapse is deterministically triggered when the resource stock is fully depleted (i.e., $R(t) \leq 0$). It may also be triggered by a stochastic existential hazard, which we model as a Bernoulli process. At each time step t, a uniform deviate $U(t) \sim U(0,\ 1)$ is drawn. If $U(t) < h$, where $h \in [0, 1]$ is the hazard rate specific to the scenario, then a collapse is triggered independently of resource levels.

Upon collapse, the system state is reset according to

$$\begin{aligned} T_{t+1} &= c_f \, T_t, \\ R_{t+1} &= 0. \end{aligned} \tag{2}$$

The system then remains inactive for r_d time steps, after which resource availability is restored as

$$R_{t+r_d} = r_f \, R_0. \tag{3}$$

We define a civilization as being ON when it maintains a positive level of resources (i.e., $R(t) > 0$). This assumption reflects the idea that some minimal material throughput is necessary for societal function and detectability, including energy consumption, communication, and technosignature production. Conversely, when resources are fully depleted ($R(t) = 0$), the civilization is considered OFF, representing a dormant, collapsed, or otherwise undetectable state. This binary definition underlies the calculation of duty cycle and related metrics discussed in subsection 3.3.

3.2 Scenario typologies and parameter mapping

We apply our model to the ten future scenarios to simulate the long-term behavior of civilizations under diverse sociotechnical conditions. Each scenario describes a distinct vision for humanity's trajectory, defined by combinations of resource abundance, governance structure, institutional resilience, and technological integration. Our goal is to translate these qualitative narratives into a coherent set of quantitative parameters that govern dynamics of growth, collapse, and recovery. Scenarios are characterized along three conceptual axes: resource regime (scarcity vs. nonscarcity), governance type (from autocratic to participatory), and structural design (hierarchical vs. distributed). Scenarios are also mapped onto six thematic clusters (Table 1).

We assign parameter values based on the sociotechnical features of each scenario. The growth rate r is taken directly from Haqq-Misra et al. (2025) and held fixed. We note that assuming

Table 1: Scenario descriptions and sociotechnical tags

Scenario	Tags	Description
S1: Big Brother is Watching	Scarcity + Rule by One + Hierarchical Cluster 6 (Earth 2024)	Centralized authoritarian system under resource stress
S2: Wild West	Scarcity + Rule by Few + Hierarchical Cluster 6 (Earth 2024)	Deregulated, competitive, and fragmented society
S3: Golden Age	Non-scarcity + Rule by All + Distributed Cluster 6 (Earth 2024)†	Stable egalitarian society with abundant resources
S4: Living with the Land	Scarcity + Rule by Few + Hierarchical Cluster 5 (Simplicity)	Low-impact, cyclical, ecologically constrained society
S5: Transhumanism	Non-scarcity + Rule by All + Distributed Cluster 4 (Engineering)	Technologically integrated, post-biological civilization
S6: Sword of Damocles	Scarcity + Rule by Few + Hierarchical Cluster 4 (Engineering)	High-tech and high-risk system with cascading failure potential
S7: Restoration	Non-scarcity‡ + Rule by All + Distributed Cluster 3 (Sustainability)	Recovered post-collapse world with resilience
S8: Ouroboros	Scarcity + Rule by Few + Hierarchical Cluster 3 (Sustainability)	Oscillatory civilization with cyclical collapse and regrowth
S9: Deus Ex Machina	Non-scarcity + Rule by All + Distributed Cluster 2 (Tech Escape)	Aggressive technological expansion with instability risk
S10: Out of Eden	Non-scarcity + Rule by All + Distributed Cluster 1 (Parallel Life)	Harmonized coexistence with nature and machines

†Cluster assignments follow Haqq-Misra et al. [2025a]; S3 shares the Earth 2024 technology cluster with S1 and S2 despite differing in global factors.

‡S7 is tagged non-scarcity in the original framework but parameterized with scarcity-level resource pressure to model post-collapse recovery (see text).

technological capacity increases linearly each year via r is a simplification, especially given scenarios that imply discontinuities such as collapse points or radical transformation points; we retain it here for consistency with the source parameterization. The remaining parameters (initial resource stock R_0, per-year depletion rate δ, collapse depth c_f, recovery delay r_d, recovery fraction r_f, and existential hazard rate h) are derived from the narrative structure of each scenario and converted into quantitative values through structured reasoning.

We assume that governance structure influences how technological civilizations respond to systemic stress and collapse. Highly centralized systems (rule by one) could be more vulnerable to severe and prolonged disruptions, as decision-making and critical infrastructure are concentrated and potentially less adaptable. When collapse occurs (due to resource exhaustion or external hazard) such civilizations may face steeper declines in technological capacity and slower recovery, particularly if institutional continuity is fragile. In contrast, more distributed governance (rule by all) might support greater redundancy and flexibility, enabling shallower collapses and faster recovery following disruption. To reflect these hypothesized patterns, we selected representative parameter value ranges for collapse depth c_f, recovery delay r_d, and recovery fraction r_f that align with the governance structure assumed in each scenario (Table 2). These values are not based on empirical observation, but represent internally consistent assumptions designed to explore contrasting trajectories in our scenario-based simulations.

One exception is the Restoration scenario (S7). While it is categorized as non-scarcity and governed by distributed institutions, it is conceptually framed as a rebuilding civilization emerging from an earlier collapse. To reflect this, we initialize the simulation with a reduced effective resource stock and a moderate depletion rate (parameters that would otherwise suggest scarcity), thereby modeling a society recovering from prior systemic failure. This allows us to represent S7 as a post-collapse recovery regime without forcing a collapse to occur within the simulation window.

The Ouroboros scenario (S8) is treated differently. In the original framework, S8 is described as a civilization that repeatedly undergoes expansion, collapse, and regrowth, with the year 3000 situated partway through an ongoing recovery phase. Rather than imposing externally timed or periodic collapse events, we model this behavior as an emergent dynamical regime. Specifically, S8 is assigned moderate technological growth, partial technological survival during collapse, and a relatively high recovery fraction of planetary resources. This combination allows repeated collapse–recovery cycles to arise endogenously from the interaction between resource depletion and rebuilding, and permits exploration of whether oscillatory behavior remains stable or dampens over long timescales.

Similarly, Deus Ex Machina (S9) is tagged as non-scarcity in the original framework but is parameterized with moderate resource pressure ($R_0 = 900$, $\delta = 1.3$) to reflect the high extraction demands of its aggressive technological expansion program, which drives resource depletion despite the scenario's nominally post-scarcity classification.

Scenarios also differ fundamentally in their assumptions about planetary resource conditions. Rather than define scarcity solely in terms of starting resources, we interpret it as a function of resource pressure (i.e., how quickly a civilization depletes its reserves). That is, systems with a large initial stock may still experience scarcity if it depletes resources at a high rate (e.g., S6). Conversely, a civilization with a modest stock may persist through repeated collapse–recovery cycles if post-collapse resource recovery partially replenishes its base (e.g., S8, S9), or may rebuild following an earlier systemic failure (e.g., S7).

We implement Monte Carlo simulations to explore how uncertainty in model parameters shapes

Table 2: Mapping sociotechnical structure to parameter logic

Governance Type	Collapse Depth c_f	Recovery Delay r_d	Recovery Fraction r_f
Rule by One	0.5–0.7	40–60	0.10–0.20
Rule by Few	0.2–0.6	50–100	0.10–0.35
Rule by All*	0.3–1.0	0–70	0.20–1.0

*Post-scarcity idealizations (S3, S10) use $c_f = 1.0$, $r_d = 0$, $r_f = 1.0$, reflecting no-collapse conditions. S7 and S9 use parameters outside the typical governance range as narrative exceptions (see text).

Table 3: Interpreting resource pressure from R_0 and δ

Resource Pressure	R_0	δ	Scenarios
High	≤ 1000	1.0–2.5	S1, S2, S4, S7, S8, S9
Moderate	400–1600	0.5–1.0	S6
Low	≥ 1200	0.0–0.5	S3, S5, S10

the long-term behavior of each scenario. Each scenario's baseline values (Table 4) are perturbed using probabilistic variation to capture plausible uncertainty (Table 5). Initial resource stock (R_0) and depletion rate (δ) are sampled from normal distributions with modest standard deviations to represent uncertainty in ecological baselines and consumption patterns. Collapse depth (c_f) and recovery fraction (r_f) are sampled using triangular distributions centered on scenario-specific values, allowing outcomes to concentrate near expected levels while still capturing asymmetries in resilience and recovery capacity. Recovery delay (r_d) is sampled from a normal distribution with a standard deviation of ± 10 years, reflecting variability in rebuilding pace across different societal structures.

The existential hazard rate (h) is held fixed for each scenario because it encodes an assumed background level of external risk (such as AI misalignment, interstellar conflict, or catastrophic environmental feedbacks) rather than internal system dynamics. Unlike other parameters whose variability arises from structural or operational uncertainty, h reflects a narrative-level judgment about a civilization's exposure to exogenous threats. Scenarios characterized by unstable geopolitical contexts, risky technological dependencies, or unresolved existential challenges (e.g., S1, S6, S7) are assigned higher hazard rates to reflect this vulnerability, while S5 carries a small residual risk due to high systemic complexity. In contrast, scenarios with societal stability, low-impact lifestyles, or resilient infrastructure (e.g., S3, S4, S8, S9, S10) are assigned zero or near-zero hazard rates. Keeping h fixed rather than sampling it probabilistically allows for clearer attribution of collapse outcomes to either endogenous fragility or exogenous shocks, rather than conflating the two.

Table 4: Final parameter values for each scenario

Scenario	r	R_0	δ	c_f	r_d	r_f	h
S1	0.0020	400	1.2	0.6	40	0.15	0.003
S2	0.0026	700	1.5	0.3	70	0.10	0.001
S3	0.0001	2400	0.0	1.0	0	1.0	0.0
S4	0.0060	600	2.5	0.2	100	0.10	0.0
S5	0.0003	2500	0.1	0.9	20	0.80	0.0005
S6	0.0018	1600	1.0	0.4	50	0.25	0.005
S7	0.0060	400	1.5	0.3	60	0.20	0.001
S8	0.0015	800	1.3	0.6	70	0.35	0.0
S9	0.0011	900	1.3	0.5	70	0.35	0.0
S10	0.0002	2700	0.0	1.0	0	1.0	0.0

Table 5: Parameter uncertainty assumptions for Monte Carlo sampling

Parameter	Distribution	Uncertainty
R_0	Normal	$\pm 5\%$ of mean
δ	Normal	$\pm 5\%$ of mean
c_f	Triangular	$\pm 5\%$ of mode
r_d	Normal	± 10 years
r_f	Triangular	$\pm 5\%$ of mode

3.3 Metrics

We define four metrics to reflect different aspects of technospheric persistence and collapse dynamics across Monte Carlo simulations. These metrics are designed to capture the onset, frequency, and continuity of civilization-level activity, with a focus on their implications for detectability.

The first metric, T_c, is the mean time to first collapse across replicates. It serves as a proxy for system fragility: scenarios with low values of T_c tend to break down early, while those with higher values demonstrate greater initial stability. When a simulation run does not experience collapse within the 1000-year window, the full timespan is used. The second metric, f_{nc}, denotes the fraction of runs that avoid collapse entirely. This value reflects the overall robustness of a scenario and helps differentiate consistently stable futures from those vulnerable to systemic failure. We also measure D_c, the average duty cycle, defined as the fraction of time a civilization remains active, that is, maintaining nonzero resource levels and thus potentially emitting technosignatures. This definition assumes that minimal resource availability is a prerequisite for detectable activity. Scenarios with high D_c maintain persistent activity, while those with low D_c may cycle through long dormant periods or experience early termination. Finally, N_c, the mean number of collapse-recovery cycles, captures whether a scenario tends to exhibit singular collapse events or undergo repeated disruptions over time. Together, these metrics provide a compact but informative summary of each scenario's trajectory, enabling comparisons across sociotechnical regimes and offering insight into the reliability and intermittency of technosignature emission.

4 Results

4.1 Monte Carlo Outcomes and Temporal Profiles

We ran 200 independent simulations for each scenario, introducing probabilistic variation in model parameters as described in Section 3. Figure 1 shows the resulting trajectories of global technology $T(t)$ and available resources $R(t)$, averaged across runs. These profiles reveal the distinct collapse and recovery dynamics that characterize each scenario. As expected, Golden Age (S3) and Out of Eden (S10) show uninterrupted growth and resource stability throughout the 1000-year window, consistent with their post-scarcity narratives and reflected in perfect duty cycles ($\overline{D}_c = 1.0$) and no collapse events ($\overline{N}_c = 0$). In stark contrast, scenarios such as Big Brother (S1) and Sword of Damocles (S6) suffer early collapses ($\overline{T}_c < 210$) and high collapse frequency ($\overline{N}_c \approx 4.5$ for S6 and nearly 10 for S1), leading to sharp contractions in technological capacity and prolonged periods of inactivity. Deus Ex Machina (S9), although collapsing later ($\overline{T}_c \approx 689$), also experiences multiple interruptions and never recovers full continuity, with only modest duty cycle ($\overline{D}_c \approx 0.91$) and about 1.5 collapse events on average.

More complex dynamics emerge in intermediate cases. Transhumanism (S5) avoids collapse in roughly two-thirds of runs and achieves a near-continuous duty cycle ($\overline{D}_c \approx 0.99$), albeit with slower growth and occasional extinction. Living with the Land (S4), in contrast, undergoes frequent and deep collapses ($\overline{N}_c \approx 6.6$), producing the lowest sustained activity among all non-terminal scenarios ($\overline{D}_c \approx 0.38$) despite a steep initial growth phase. Restoration (S7), designed as a post-collapse recovery regime, displays consistent rebuilding behavior: early collapses ($\overline{T}_c \approx 223$), followed by approximately 7.5 recovery cycles and moderate continuity ($\overline{D}_c \approx 0.57$). Ouroboros (S8) exhibits a distinct pattern of low-frequency, self-stabilizing oscillations: it collapses only twice on average, yet maintains robust technological levels and a high duty cycle ($\overline{D}_c \approx 0.87$), reflecting a regime that cycles gracefully without exhausting resources or triggering runaway collapse.

Because the S8 narrative implies repeated collapse–recovery cycles, we ask whether those oscillations are always transient or can be sustained; we probe this with a sweep over r_f and δ (Figure 2). Panel A shows deterministic technology trajectories for varying r_f while holding all other S8 parameters at their baseline values. For low recovery fractions ($r_f \leq 0.15$), the system undergoes rapid, repeated collapse–recovery cycles (3 cycles within the 1000-year window), as the diminished resource base after each recovery is quickly exhausted. At intermediate values ($0.20 \leq r_f \leq 0.50$), the system sustains two collapse–recovery cycles with longer growth phases between collapses. At higher recovery fractions ($r_f = 0.60$), the first collapse is delayed significantly, and only a single cycle occurs within the simulation window. Panel B maps the number of collapse cycles across the (r_f, δ) parameter space, revealing a clear regime boundary: high depletion rates combined with low recovery fractions produce the most frequent oscillations (up to 8 cycles), while low depletion and high recovery fractions suppress collapse entirely. The baseline S8 parameters (marked with a star) sit near the boundary between 2- and 3-cycle regimes, suggesting that modest shifts in resource recovery capacity could qualitatively alter the civilization's long-term trajectory. These results confirm that S8's oscillatory behavior is not universally stable and it depends sensitively on the balance between resource depletion and post-collapse recovery, with some parameter combinations producing sustained cycling and others leading to dampened or terminal collapse.

Figure 3 summarizes the outcomes of these simulations using the average of the four metrics described above ($\overline{T}_c$, f_{nc}, $\overline{D}_c$, $\overline{N}_c$). While Golden Age (S3) and Out of Eden (S10) remain fully stable throughout the 1000-year window, with no collapses and continuous activity ($f_{nc} = 1.0$,

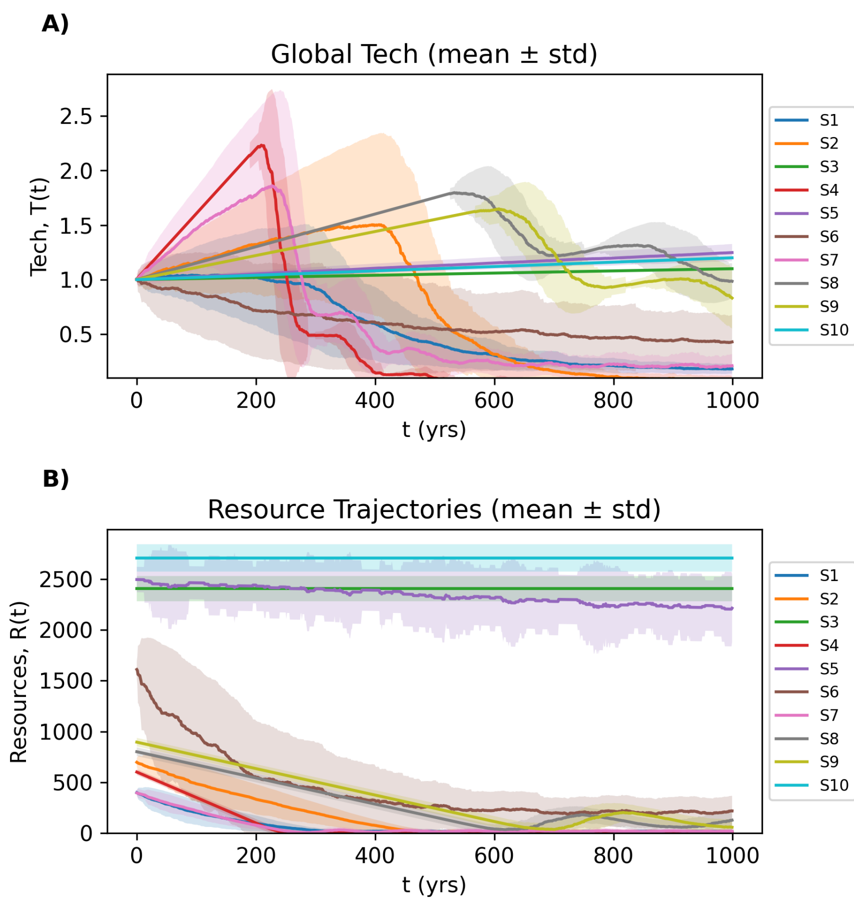

Figure 1: Temporal dynamics of simulated civilizations across ten scenarios. (A) Mean trajectories of global technology $T(t)$ with ± 1 standard deviation shaded. (B) Mean resource levels $R(t)$ with ± 1 standard deviation shaded.

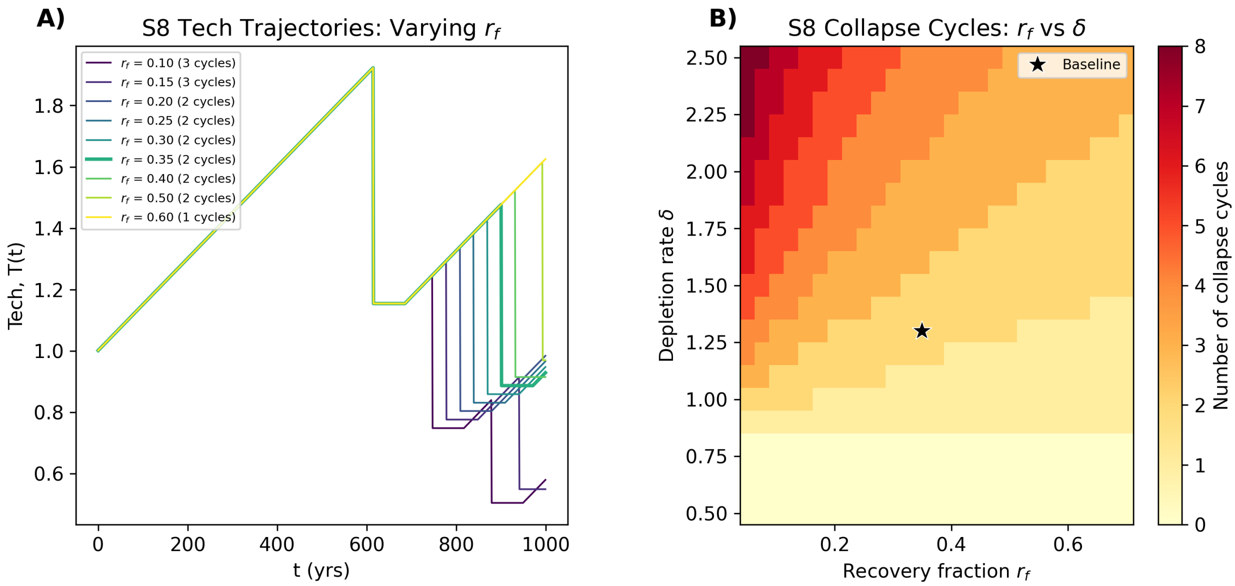

Figure 2: Parameter sensitivity of S8 (Ouroboros) oscillatory dynamics. (A) Deterministic technology trajectories $T(t)$ for varying recovery fraction r_f, with all other parameters held at baseline values. The number of collapse–recovery cycles within the 1000-year window is indicated in the legend. (B) Number of collapse–recovery cycles as a function of recovery fraction r_f and depletion rate δ. The star marks the baseline S8 parameter combination.

$\overline{D}_c = 1.0$), other futures display more volatile dynamics. Big Brother (S1) and Sword of Damocles (S6) show early collapses ($\overline{T}_c \leq 220$) and high collapse frequency, while Deus Ex Machina (S9) collapses later ($\overline{T}_c \approx 689$) but still undergoes multiple recovery cycles, reflecting fragile systems with intermittent technosignature potential.

4.2 Effective Detectability and Duty Cycle Analysis

Our simulations reveal that many plausible civilizational futures exhibit intermittent collapse and recovery cycles, resource-driven dormancy, or stochastic failure due to external hazards. These patterns imply that civilizations may oscillate between ON and OFF phases, producing irregular technosignature signals that are easily missed. To account for this, we introduce an effective detectability duration,

$$L_{\text{eff}} = D_c T_{\text{span}}, \tag{4}$$

where D_c is the mean duty cycle (defined above as the fraction of time a civilization remains active) and T_{span} is the observation window (1000 years in our simulations). Some have also proposed a related refinement to Drake's equation in the form of a "reappearance factor" for planets, where civilizations can arise multiple times in succession [Bhattacharya et al., 2011]. For example, even if one technological society collapses after 10,000 years, life may persist on that planet and eventually give rise to a new intelligent species. This factor differs from duty cycle in that it counts sequential independent civilizations over geological timescales, whereas duty cycle refers to the active vs. inactive phases within a single civilization's lifespan.

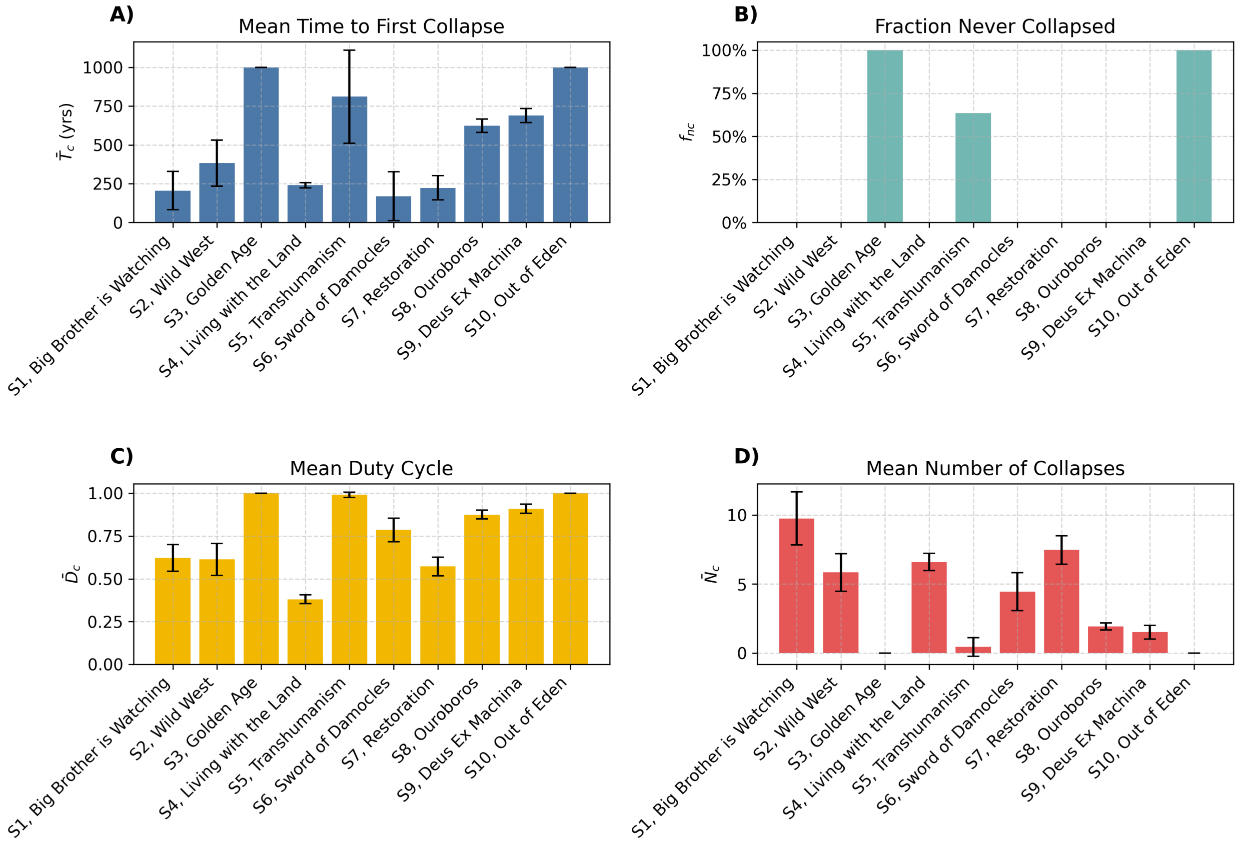

Figure 3: Distributions of key collapse and recovery metrics across scenarios (200 runs each). (A) Mean time to first collapse $\overline{T}_c$ with standard deviation bars. (B) Fraction of simulations that never experienced collapse f_{nc}. (C) Mean duty cycle $\overline{D}_c$, defined as the fraction of time the technosphere remains active. (D) Mean number of collapse events $\overline{N}_c$.

Among the ten scenarios, D_c spans a wide range in our simulations. In the most collapse-prone scenario, S4, D_c was as low as 0.381, indicating these civilizations spend well over half of their time in a collapsed or dormant state. By contrast, scenarios S3 and S10 had $D_c = 1.00$, never experiencing a collapse (continuous activity). Most scenarios fall somewhere in between. For example, S5 has a high $D_c \approx 0.99$, whereas S2 reaches only $D_c \approx 0.61$, reflecting prolonged instability. This variability in D_c has direct implications for the detectability of civilizations. Even if intelligent life is widespread, the duty cycle reduces the probability of intercepting a signal at a given moment. Civilizations may exist for millennia, but if they emit only intermittently, the chances of overlap with our own observational window shrink significantly.

The simulation results also raise a deeper question about the distribution of civilizations in the galaxy: should we expect them to be uniform in character and behavior, or drawn from a more diverse ensemble of trajectories? A uniform distribution, where all civilizations resemble a stable high-duty scenario like S3, would imply a galaxy populated by long-lived, continuously emitting technospheres. Such a distribution would make the silence we observe profoundly puzzling. On the other hand, if the distribution is non-uniform, with most civilizations following fragile, collapse-prone paths (e.g., S1, S2, S6), then long-duty-cycle civilizations would be statistical outliers. In this case, silence becomes more plausible, not because civilizations are rare, but because most are quiet or transient for the vast majority of their lifespan.

Our findings support the latter view. Out of 200 simulations per scenario, only S3 and S10 avoided collapse entirely ($f_{nc} = 1.00$). Most other scenarios saw high collapse frequency (e.g., $\overline{N}_c = 9.76$ for S1), and moderate or low duty cycles. These collapse-prone scenarios suggest that many civilizations may not follow monotonic energy-growth trajectories. Instead of ascending smoothly up the Kardashev scale, they may oscillate or even regress, casting doubt on the assumption that energy use (and thus detectability) increases predictably over time. This indicates that robustness might not be the norm. Civilizations that achieve continuous detectability, such as those in post-scarcity or ecologically harmonized futures, appear rare and structurally exceptional. If galactic civilizations follow a similar distribution, then the Great Silence may simply reflect the dominance of fragile or intermittently silent technospheres.

In this light, the silence we observe may not contradict the possibility of intelligent life, rather, it may reflect the statistical weight of civilizations that emit briefly or not at all during our observational epoch. A non-uniform distribution with many S1/S2-type worlds and very few S3/S10-type worlds reconciles the Drake Equation with current data, while supporting the need for SETI strategies that target not just strong, continuous signals but also faint, episodic, or relic technosignatures.

4.3 Persistence of Technosignatures Beyond Civilizational Activity

The effective detectability term we propose, L_{eff}, captures the active fraction of a civilization's technological lifespan (i.e., the periods when it is ON and producing observable outputs). However, this formulation assumes that detectability ends when activity ceases. In practice, many technosignatures may outlast their creators, continuing to be observable well after the civilization itself has collapsed or gone quiet. To account for this, we introduce a persistence term, τ, representing the average duration that technosignatures remain detectable after each *ON* period. The revised detectability window, $L_{\text{eff}}^{(d)}$, becomes

$$L_{\text{eff}}^{(d)} = D_c\, T_{\text{span}} + N_{\text{ON}}\, \tau. \tag{5}$$

where N_{ON} is the number of active intervals in the simulation. This extension is particularly relevant for short-lived or intermittent civilizations as long-lasting residual signals could greatly increase detectability even if active periods are brief.

Orbital debris and artificial satellites can last from decades to as long as 10,000 to 100,000 years. Satellites in low Earth orbit typically deorbit within decades due to atmospheric drag, while those in geosynchronous orbit, facing minimal resistance, may persist for much longer. Without maintenance, a dense band of satellites in the so-called Clarke belt would slowly degrade through collisions and orbital perturbations over tens of thousands of years, eventually becoming undetectable [Socas-Navarro, 2018]. Megastructures such as Dyson spheres, space mirrors, or stellar engines may persist longer still, though their survival depends on structural design and orbital stability. Rigid constructs without active station-keeping may collapse or drift within a few centuries to several thousand years [Wright, 2020], however, more modular configurations, like swarms of solar collectors, could remain intact for up to hundreds of thousands or even millions of years if not disrupted by collisions.

Atmospheric technosignatures provide the shortest windows of detectability. On Earth, CFC-11 and CFC-12 have atmospheric lifetimes of about 55 and 140 years, respectively, before being broken down by photolysis or other chemical processes [Unk]. Other compounds are more resilient: tetrafluoromethane (CF_4) persists for ~50,000 years, hexafluoroethane (C_2F_6) for ~10,000 years, and octafluoropropane (C_3F_8) for ˜2,600 years [Trudinger et al., 2016]. Longer-lived isotopic signatures are conceivable, but their remote detection remains technically challenging. Electromagnetic technosignatures such as radio or laser transmissions propagate indefinitely but are only practically detectable for finite periods. Focused, high-power signals might remain observable for a few hundred light-years over spans of centuries to millennia, depending on the strength of the signal and the sensitivity of receivers [Sheikh et al., 2025].

These persistence times imply that τ can range from negligible (radio bursts) to dominant (orbital infrastructure). In the case of short-lived civilizations with brief ON phases, the contribution of $N_{\mathrm{ON}} \cdot \tau$ may exceed that of $D_c \cdot T_{\mathrm{span}}$. In contrast, for continuously active or long-lived civilizations, L_{eff} dominates and τ contributes relatively little. This distinction matters. If most detectable technosignatures are short-lived (e.g., radio leakage), then D_c and T_{span} remain the key parameters, and our results suggest low duty cycles make detection unlikely. But if many technosignatures persist for millennia beyond collapse, then the silence we observe becomes harder to attribute solely to civilizational fragility. Instead, it would require that either technosignatures are intrinsically ephemeral, or civilizations rarely produce persistent artifacts.

5 Discussion

5.1 Reframing the Drake Equation's Longevity Term

Our results reframe the Drake Equation's longevity term in a way that addresses the Fermi Paradox with greater nuance. Rather than requiring that intelligent life be exceedingly rare, the Great Silence may simply indicate that technosignatures are intermittent and timing-dependent [Balbi, 2018]. Aligned with this, recent work suggests that civilizations might commonly face phase transitions or limits to growth that prevent long, unbroken periods of signal emission [Wong and Bartlett, 2022]. In fact, our bimodal outcomes, with some scenarios yielding repeated collapse and others achieving long-term stability, closely echo this hypothesis. The overall idea is that planetary civilizations may follow one of two paths: either "asymptotic burnout", collapsing under the weight of exponential growth, or a "homeostatic awakening" in which they deliberately curb expansion to achieve equilibrium. In the latter case, a civilization might survive indefinitely but become hard to detect due to a lack of aggressive expansion or high-power signaling. Our high duty-cycle scenarios like Golden Age (S3) or Out of Eden (S10) resemble this stable, sustainable type, whereas the collapse-prone scenarios like Big Brother (S1) or Sword of Damocles (S6) are akin to burnout cases. The implication is that the rarity of continuous technosignature emitters (high L or D_c) in our simulations could reflect a Galactic trend, where long-lived, broadcasting civilizations might be the exception, not the rule.

To formalize this connection, we recall the Drake Equation [Drake, 1965], which estimates the number of detectable civilizations in our galaxy as

$$N = R_* \, f_p \, n_e \, f_l \, f_i \, f_c \, L. \tag{6}$$

Each parameter represents a step in the emergence of detectable life: R_* is the rate of star formation, f_p the fraction of those stars with planets, n_e the number of habitable planets per system, f_l the fraction of planets where life arises, f_i the fraction where intelligence evolves, f_c the fraction that develop detectable technology, and L the average duration such civilizations remain detectable. While early terms are increasingly constrained by astrophysical observation, the final term L carries outsized importance and remains deeply uncertain. If civilizations exist but are short-lived or frequently silent, N could remain small despite favorable conditions for life elsewhere.

An important extension of this idea is that technology can outlast its creators. Our study has focused on the active phase of civilizations (the ON periods), but one must consider that even a civilization which collapses (goes OFF) might leave behind artifacts or long-lived technosignatures. For example, it has been argued that the maximum longevity of a technosignature could be far greater than the civilization's biological lifespan, because machines, probes, or infrastructure can continue operating (or at least remain observable) for millennia or longer [Wright et al., 2022]. In this view, Earth and humanity may be poor guides to the true value of L, since advanced extraterrestrial technology might persist well beyond the civilization's demise. This is a crucial point for SETI as even extinct or dormant civilizations could be detectable if they've built something durable [Ćirković et al., 2019]. Dyson's classic idea of searching for waste heat from starlight-absorbing megastructures is a case in point. A "Dyson Sphere" could remain in orbit around a star for eons, glaring in the infrared, even if the society that built it is long gone [Dyson, 1960]. Such relic technosignatures (sometimes called "Dysonian" technosignatures) broaden our notion of detectability. In the same vein, a dense belt of artificial satellites around a planet (a "Clarke exobelt") might be detectable via its subtle transit signature [Socas-Navarro, 2018]. Other examples include planetary atmospheric anomalies (e.g., chlorofluorocarbon pollution or unusual isotopic ratios in a planet's spectrum) that could signal past industrial activity [Wright et al., 2022]. In short, not all technosignatures are "live broadcasts". Some might be fossils or afterglows of past epochs of activity.

Our simulations suggest that the traditional interpretation of the Drake Equation's longevity term, L, as a continuous span of technosignature emission, may be unrealistic. Across a wide range of plausible futures, we find that civilizational activity is frequently interrupted by collapse, resource depletion, or other disruptions. In most scenarios, the emission of technosignatures is not constant but episodic, raising the need for a revised formulation. We propose an effective detectability duration, $L_{\text{eff}} = D_c \times T_{\text{span}}$, where D_c is the mean duty cycle and T_{span} is the time horizon. This redefinition captures the operational reality of civilizations that switch between active and dormant states. This intermittency reframes how N, the number of detectable civilizations, should be interpreted. If L is better understood as L_{eff}, then models that assume continuous detectability may significantly overestimate N. Conversely, incorporating duty cycle effects helps align theoretical predictions with empirical null results. A galaxy populated by civilizations with low duty cycles would remain largely silent, even if intelligent life is common.

This framework has two important implications. First, it helps explain the Great Silence. Civilizations may exist in abundance, but if they are detectable only intermittently, the likelihood that their technosignatures overlap with our current observational window is low. The apparent absence of signals may not reflect the rarity of extraterrestrial technology, but rather the rarity of sustained emission. Second, it suggests that long-lived, continuously emitting civilizations are outliers. Most of our simulated futures spent significant portions of time in silent or collapsed states, even if they eventually recovered. If such patterns are typical across the galaxy, then detection becomes a matter of timing: for two intermittent civilizations, the probability of mutual visibility is the product of their duty cycles, which is potentially very small. This stands in contrast to the assumptions embedded in the Kardashev scale [Kardashev, 1964], which classifies civilizations by their continuous energy use and implies long-term, uninterrupted detectability. Our simulations instead point to dynamic energy throughput, with civilizations temporarily reaching higher levels, such as approaching Type I, only to regress during periods of collapse or dormancy. These fluctuations reduce their cumulative detectability and challenge the static expectations of the Kardashev framework. Moreover, while our models focus on collapse-driven intermittency, it's worth noting that some civilizations might limit their energy use deliberately (in pursuit of ecological balance or long-term stability, for example), remaining below Type I not due to collapse, but by design.

Our results also help reconcile seemingly contradictory perspectives on civilizational longevity. One view posits that technological civilizations are fundamentally fragile and prone to self-destruction soon after emergence, driven by ecological, behavioral, or structural instabilities [Vinn, 2024]. Another suggests that once civilizations surpass key thresholds of resilience and coordination, they may persist indefinitely, steadily accumulating across cosmic time [Grinspoon, 2009]. Both patterns emerge naturally in our simulations. Scarcity-driven, hierarchically governed scenarios frequently enter collapse-recovery cycles, resulting in low duty cycles and limited detectability. In contrast, post-scarcity, distributed scenarios maintain stable resource use and continuous activity, supporting high-duty-cycle technospheres. These differences arise not from exogenous shocks, but from internal sociotechnical structure.

It is worth noting, however, that the Drake Equation's terms beyond f_l carry increasingly problematic hidden assumptions that our duty-cycle framework helps to expose [Cirković, 2004]. For the product $f_l \times f_i$ to be pragmatically sound, f_i must represent the fraction of life-bearing planets that develop intelligent life *at least once*, not merely once, acknowledging that intelligence may arise multiple times on a single world. Consequently, f_c must be understood as the fraction of planets with at least one instance of intelligent life on which at least one instance of traceable technology emerges. Most critically, L must refer to the duration of traceability, decoupled from the longevity of the technology's producers and its originating technology. That is, L must also account for the temporal dynamics of remnants left by extinct intelligent builders on or around worlds where new kinds of intelligent builders may not even be producing traceable technologies—a situation of spatial co-existence without temporal co-existence of technology-capable beings. In principle, multiple kinds of intelligent builders could produce different technologies with different degrees of traceability, or technologies that are even antagonistic to one another, although this complicates the picture considerably. Extending the duty-cycle metaphor, the ON-OFF behavior of a technosphere may not be "gated" solely by triggers of collapse as a rectangular pulse. Instead, it may represent the reaching and exceeding of a particular detectability threshold—contingent on the observing civilization—through the intentional or unintentional "constructive" or "destructive" "interference" of particular technologies deployed by a single civilization of builders or by multiple spatially co-existent civilizations that may or may not co-exist temporally. For the particular case of the ten scenarios modeled in this paper, we have assumed that humans are the only technology-capable species on Earth, so the issue of multiple kinds of builders does not arise; only formerly collapsed incarnations of human civilization are relevant.

Recent work by Balbi and Grimaldi [2024] provides further theoretical support for the importance of the duty-cycle framework. Applying Lindy's law to technosignature longevities, they show that for energy-intensive technoemissions the longevity distribution follows a power law $\rho_L(L) \propto L^{-\alpha}$ with $\alpha > 1$, strongly disfavoring long-lived technosignatures. Under realistic assumptions ($\alpha \geq 2$), the first detected technosignature is predicted to be relatively short-lived ($L < 10^2$ years at 95% confidence), rather than the 10^9-year signals sometimes assumed in optimistic SETI scenarios. This result strengthens the case for our framework: if most detectable technosignatures are intrinsically short-lived, then the combination of Lindy-distributed longevities and low duty cycles (as observed in our collapse-prone scenarios) implies a doubly diminished probability of detection, offering a complementary explanation for the Great Silence.

5.2 Duty Cycles of Technological Remnants

An additional consideration that extends the duty-cycle framework concerns the autonomous behavior of technological remnants. When a civilization collapses, not all of its technological infrastructure necessarily ceases to function. Some systems—particularly those designed for autonomous operation—may continue to operate on their own duty cycles, independent of the civilization that created them. For example, a planetary defense system designed to deflect asteroids might be activated and deactivated at predetermined intervals or in response to triggering events, persisting long after the civilization that built it has collapsed and while a recovering population has no memory of its existence. Similarly, automated beacons, environmental monitoring stations, or orbital maintenance systems could continue cycling through active and dormant states according to their original programming or in response to environmental triggers.

This creates a layered duty-cycle structure: the remnant technology's ON-OFF cycle is superimposed on the civilization's overall collapse-recovery cycle. During periods when the civilization is OFF (collapsed or recovering), remnant technologies in their ON phase could produce detectable technosignatures, effectively extending the civilization's observable presence beyond its active periods. Conversely, when the civilization recovers and re-enters its ON phase, it may encounter and reactivate dormant remnant systems, creating a form of constructive interference between the two duty cycles. This nuance modifies the persistence term τ in our extended formulation ($L_{\text{eff}}^{(d)} = D_c\, T_{\text{span}} + N_{\text{ON}}\, \tau$): rather than representing a simple passive decay time, τ may itself contain active intervals during which remnant technologies produce detectable signatures. The effective detectability window thus becomes the union of the civilization's active periods and the remnant technology's active periods, potentially increasing the overall duty cycle beyond what civilizational activity alone would suggest.

5.3 Limitations and Modeling Considerations

Finally, several limitations warrant discussion. Our modeling framework is grounded in Earth-originating scenario typologies, drawing from narrative futures methodologies informed by contemporary geopolitical, ecological, and technological trends. These projections embed assumptions about governance structures, resource constraints, and recovery dynamics that may not extend to non-terrestrial civilizations or to those that do not follow Earth-like developmental pathways. While this grounding enables empirical plausibility, it limits extrapolation beyond our biosphere's trajectory [Rubin, 2001, Denning, 2011]. Moreover, our simulations assume bounded planetary systems and do not account for galactic colonization, interstellar migration, or modes of expansion that could alter the spatial and temporal visibility of civilizations [Walters et al., 1980, Papagiannis, 1983]. We also exclude the effects of cultural convergence, information saturation, or technological transcendence (e.g., the emergence of post-biological intelligence), all of which could suppress or qualitatively change technosignature output. Our operational definition of detectability (based on internal sociotechnical continuity) does not differentiate among specific technosignature classes (e.g., radio emissions, waste heat, megastructures), nor does it address detection thresholds or instrumental sensitivity.

Additionally, our assumption that technological capacity increases linearly each year according to a fixed growth rate r is a significant simplification. The many discontinuities implied by the scenarios—not only collapse points but also radical transformation points such as the emergence of artificial general intelligence or breakthroughs in energy harvesting—suggest that a linear growth model may not adequately capture the dynamics of technological change across all regimes. Moreover, the available resource stock R_0 is treated as an exogenous parameter, but in practice it is contingent on the technological capacity to identify and utilize resources: a civilization that develops asteroid mining or nuclear fusion effectively increases its accessible resource base, fundamentally altering the growth-depletion dynamics. Recent work by Haqq-Misra et al. [2025b], which shares the same ten-scenario framework employed here, reframes the Kardashev scale as a luminosity limit constrained by thermodynamic efficiency (exergy) and explores exploration-exploitation trade-offs for long-lived technospheres. Their framework highlights that growth trajectories depend on whether a civilization pursues spatial expansion (exploration) or intensive resource use within a fixed domain (exploitation), a distinction that our linear growth model does not capture. Future work could incorporate exergy constraints and non-linear growth functions informed by their luminosity-limit model, as well as feedback between technological capacity and accessible resource stocks, to produce more physically grounded projections of civilizational trajectories.

In summary, our results can partially explain the Great Silence: if detectability is often sporadic, then long periods without contact may be the expected baseline. Intermittency may be typical and recognizing that could refine our expectations and expand the scope of SETI.

6 Implications for Earth's Future

While the duty-cycle framework developed in this paper is motivated by the search for extraterrestrial technosignatures, its most immediate relevance may be to Earth's own civilizational trajectory. The ten scenarios modeled here are not abstract constructs; they are projections of plausible futures for human civilization, grounded in present-day sociotechnical conditions. The collapse-recovery dynamics, duty cycles, and effective detectability durations computed from our simulations therefore offer a quantitative lens through which to evaluate humanity's own resilience and long-term prospects.

A key insight from our modeling is that the parameters governing recovery—the recovery fraction r_f and the recovery delay r_d—are not fixed properties of a civilization but can be deliberately engineered. Investments in civilizational continuity infrastructure directly map onto these parameters: a higher r_f corresponds to greater preservation of knowledge, technology, and resources through collapse events, while a lower r_d corresponds to faster institutional and technological rebuilding. Together, improvements in these parameters increase the duty cycle D_c and thus the effective detectability duration L_{eff}, but more importantly for Earth's purposes, they increase the fraction of time civilization remains functional and productive.

The concept of "recovery kits"—curated repositories of essential knowledge designed to accelerate post-collapse rebuilding—represents one practical approach to raising r_f. Dartnell [2014] has explored in detail the minimum knowledge base required to reconstruct industrial civilization from scratch, spanning agriculture, materials science, medicine, power generation, and communication. Such recovery kits, whether physical or digital, function as a form of civilizational insurance: they ensure that the knowledge accumulated over centuries of development is not lost when institutional continuity is broken.

Parallel efforts in biological resource preservation serve a complementary function. The Svalbard Global Seed Vault, which now holds over one million seed samples from nearly every country on Earth [Asdal and Guarino, 2020], preserves agricultural biodiversity against civilizational disruption, ensuring that post-collapse societies retain access to the genetic resources needed to rebuild food systems. More recently, Colossal Biosciences has established a BioVault in Dubai dedicated to preserving broader biodiversity through cryopreservation of cellular material from threatened species [Colossal Biosciences]. These vaults represent real-world implementations of resilience infrastructure that, in the language of our model, serve to increase r_f by preserving critical biological and genetic capital through collapse events.

Viewed through the duty-cycle framework, the question of civilizational resilience becomes quantifiable: what is the marginal improvement in D_c from a given resilience investment? Our parameter sensitivity analysis of the Ouroboros scenario (S8) demonstrates that modest shifts in the recovery fraction r_f can qualitatively alter the long-term trajectory of a civilization, transitioning it from frequent, damaging collapse cycles to sustained or dampened oscillations. This suggests that targeted investments in knowledge preservation, resource vaults, distributed energy systems, and redundant communication networks could yield disproportionate returns in civilizational continuity. The duty-cycle framework thus provides not only a tool for interpreting the Great Silence, but also a practical metric for evaluating the long-term value of resilience planning on Earth.

7 Conclusions

We have developed a hybrid deterministic-stochastic model to simulate collapse-recovery dynamics across ten plausible futures for Earth-originating civilization, and used these simulations to quantify the duty cycles and effective detectability durations of technological civilizations. Our principal findings are as follows.

First, collapse-recovery dynamics produce a wide range of duty cycles across plausible civilizational futures. Among the ten scenarios, the mean duty cycle D_c spans from approximately 0.38 (Living with the Land, S4) to 1.00 (Golden Age, S3; Out of Eden, S10), with most scenarios exhibiting significant periods of dormancy or collapse. This variability has direct implications for the probability of detecting extraterrestrial technosignatures at any given moment.

Second, the effective detectability duration $L_{\text{eff}} = D_c T_{\text{span}}$ provides a more realistic replacement for the traditional Drake Equation's static longevity term L. By incorporating the fraction of time a civilization is actually active, this formulation accounts for the intermittent nature of technosignature emission and helps reconcile theoretical predictions with the observed absence of extraterrestrial signals. When combined with recent work suggesting that technosignature longevities follow power-law distributions disfavoring long-lived signals [Balbi and Grimaldi, 2024], the duty-cycle framework implies a doubly diminished probability of detection for most civilizations.

Third, technological remnants from collapsed civilizations may exhibit their own ON-OFF dynamics, creating layered duty-cycle structures that extend the effective detectability window beyond active civilizational periods. This nuance modifies the persistence term in our extended formulation and suggests that even dormant civilizations may intermittently produce detectable signatures through autonomous remnant systems.

Fourth, the duty-cycle framework has implications beyond SETI. Applied to Earth's own trajectory, it provides a quantitative lens for evaluating investments in civilizational resilience—knowledge preservation archives, genetic resource vaults, and distributed infrastructure—as interventions that raise the recovery fraction r_f and lower the recovery delay r_d, thereby increasing humanity's own duty cycle.

Several avenues for future work emerge from this study. Incorporating exergy constraints and non-linear growth functions informed by thermodynamic limits to growth [Haqq-Misra et al., 2025b] could yield more physically grounded simulations. Modeling the independent duty cycles of technological remnants and their interaction with civilizational recovery would add further realism. Finally, extending the framework to account for multiple co-existent technology-producing species or for feedback between technological capacity and accessible resource stocks would broaden its applicability to non-terrestrial contexts.

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